

Module 13: Quality of Service (QoS) and Traffic Shaping

Unit 4: Transport Layer

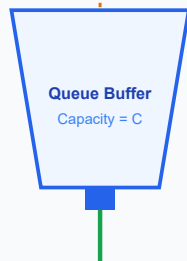
QoS & Traffic Regulation

QoS Performance Parameters, Queue Scheduling Disciplines (FIFO, Priority, WFQ), Leaky Bucket vs Token Bucket Traffic Shaping, and Burst Duration Math.

Quality of Service (QoS): Traffic Shaping & Bucket Algorithms

1. Leaky Bucket Algorithm (Traffic Smoothing)

Bursty Unpredictable Input



Constant Uniform Output Rate (r)

Core Working Principles:

- Inflow can be highly variable and bursty.
- If buffer fills completely, arriving packets are **DISCARDED**.
- Outflow rate is strictly constant (Cannot accommodate bursts!).

2. Token Bucket Algorithm (Allows Controlled Bursts)

- Tokens generated at constant rate R tokens/sec.
- Bucket accumulates up to capacity C tokens.
- Transmitting K bytes requires consuming K tokens.

Maximum Burst Duration Formula (AKTU 10-Marks):

$$\text{Burst Time: } T = C / (M - R)$$

(M = Peak Transmission Rate, R = Token Rate, C = Capacity)

4 Key QoS Performance Metrics:

- **Latency (Delay):** Time for a packet to travel source to destination.
- **Jitter:** Variation in packet arrival delay (crucial for smooth voice/video).
- **Bandwidth / Throughput:** Bits transmitted per unit time.
- **Packet Loss:** Percentage of dropped packets under buffer overflow.

Scheduling Techniques: FIFO | Priority Queuing | WFQ

DiffServ (Differentiated Services): Uses IP ToS byte for per-hop QoS.

1. Token Bucket Burst Mathematics

$$\text{Burst Duration: } T = C / (M - R)$$

$$\text{Total Transmitted Data: } D = M * T = C * M / (M - R)$$

Example: $C = 1\text{MB}$, $M = 50\text{MB/s}$, $R = 10\text{MB/s}$ -> $T = 1 / 40 = 25$ milliseconds (1.25 MB total data)

2. Leaky Bucket vs Token Bucket Comparison

Parameter	Leaky Bucket	Token Bucket
Output Nature	Rigidly constant uniform rate	Allows controlled bursts up to capacity C
Overflow Action	Discards arriving packets	Discards surplus tokens (Zero packet loss)
Suitability	Constant bit-rate streams (Audio/VoIP)	Bursty applications (HTTP web browsing, FTP)